1. Microsoft Threat Modeling

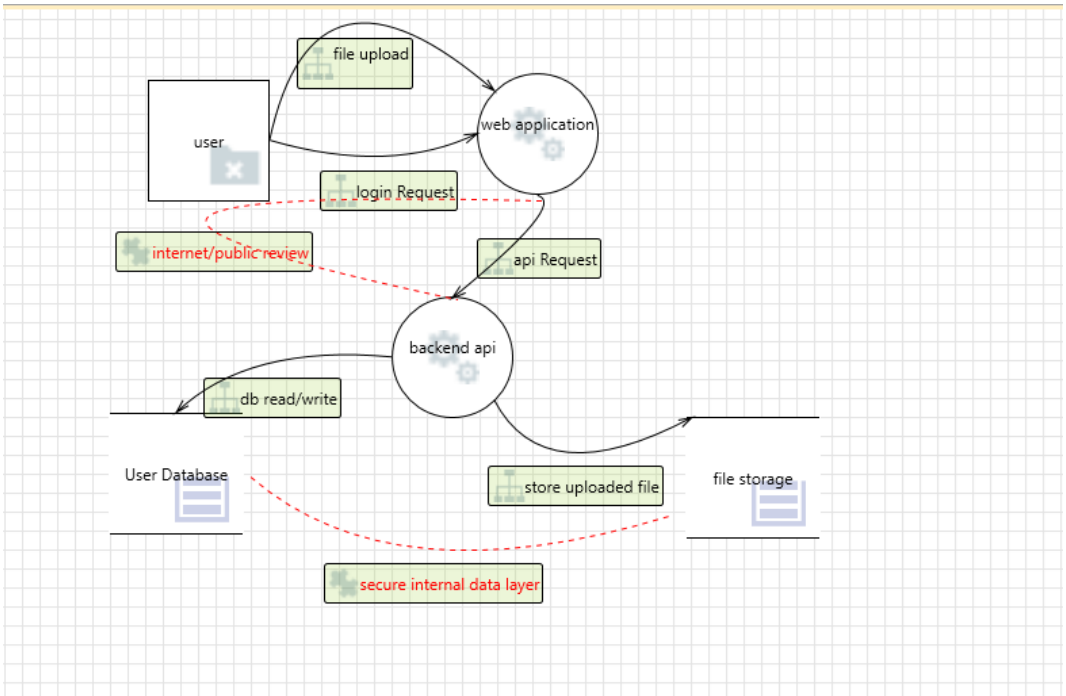
Aim

To identify potential security threats and vulnerabilities in a given application scenario using Microsoft Threat Modeling Tool, and to recommend appropriate security controls.

Procedure

1. Install and open Microsoft Threat Modeling Tool.
2. Create a new threat model.
3. Define the application scenario and its components.
4. Identify:
   - External entities
   - Processes
   - Data stores
   - Data flows
   - Trust boundaries
5. Draw the Data Flow Diagram (DFD) for the given scenario.
6. Define the properties of data flows and components.
7. Run the threat-generation/analysis feature.
8. Review the threats identified using the STRIDE methodology:
   - Spoofing
   - Tampering
   - Repudiation
   - Information Disclosure
   - Denial of Service
   - Elevation of Privilege
9. Analyze each identified threat and assign appropriate mitigation controls.
10. Generate/save the threat model report.

Output



Data Flow Diagram with trust boundaries

Result

The given application scenario was successfully modeled, potential STRIDE-based threats were identified, and suitable security mitigations were recommended.